

Assignment – 4

1. Explain the difference between `arr[:,0]` and `arr[:,[0]]` with an example.

Both are used to select the **first column** of a NumPy array, but they return different types of arrays.

- `arr[:,0]` returns a **1D array (vector)**.
- `arr[:,[0]]` returns a **2D array (matrix)**.

Example:

```
import numpy as np
arr = np.array([[1, 2, 3],
               [4, 5, 6],
               [7, 8, 9]])

print(arr[:,0])
print(arr[:,[0]])
```

Output:

```
[1 4 7]
```

```
[[1]
 [4]
 [7]]
```

Difference

`arr[:,0]`

Returns a 1D array

Shape: (3,)

Used when a vector is needed

`arr[:,[0]]`

Returns a 2D array

Shape: (3,1)

Used when a column matrix is required

2. Explain the difference between Vectorization and Broadcasting in NumPy with proper example.

Vectorization

Definition:

Vectorization means performing operations on an entire array at once without using loops. It makes the code shorter and faster.

Example:

```
import numpy as np
arr = np.array([1, 2, 3, 4])
result = arr * 5
print(result)
```

Output:

```
[ 5 10 15 20]
```

Explanation:

Each element of the array is multiplied by 5 automatically without using a `for` loop.

Broadcasting

Definition:

Broadcasting allows NumPy to perform arithmetic operations on arrays of different shapes by automatically expanding the smaller array to match the larger one.

Example:

```
import numpy as np
arr = np.array([[1, 2, 3],
               [4, 5, 6]])
b = np.array([10, 20, 30])
result = arr + b
print(result)
```

Output:

```
[[11 22 33]
 [14 25 36]]
```

Explanation:

The 1D array `[10, 20, 30]` is automatically broadcast to each row of the 2D array before addition.

Difference between Vectorization and Broadcasting

Vectorization

Performs operations on entire arrays without loops.

Improves execution speed by avoiding explicit loops.

Example: `arr * 5`

Broadcasting

Performs operations on arrays of different shapes by automatically expanding the smaller array.

Makes arithmetic between differently shaped arrays possible.

Example: `arr + [10, 20, 30]`